

Floating Point

MDE - Advanced Engineering Topics, ITESO

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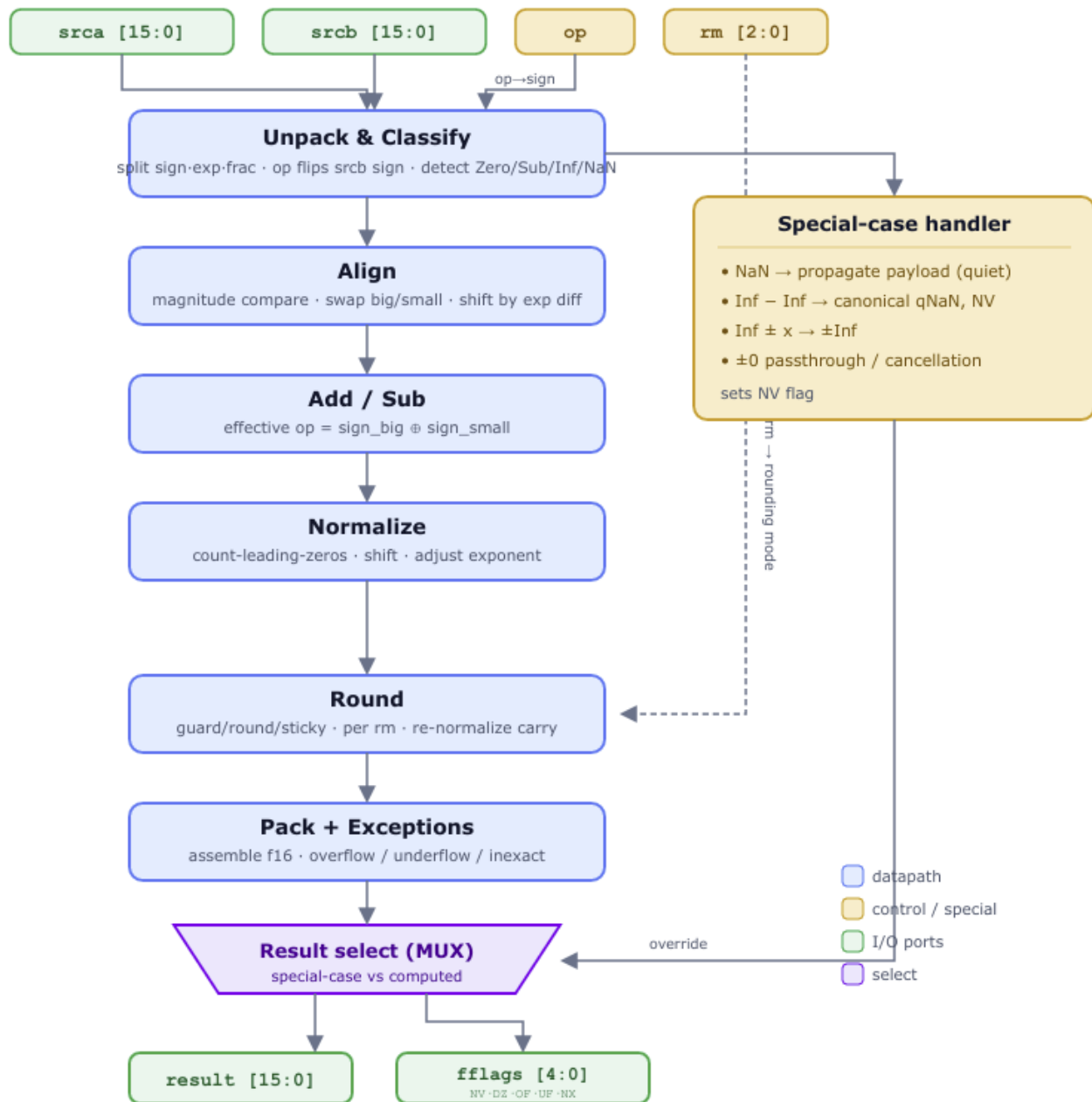
Guadalajara, Mexico

Architecture and considerations

The architecture is a half precision (16-bits) floating point adder/subtractor (IEEE 754 binary16). It is fully combinational and supports five rounding modes. An unpack stage splits each operand into sign, exponent, and fraction, restores the implicit bit, and classifies zeros, subnormals, infinities, and NaNs.

The datapath follows the classic five stages: alignment (swap by magnitude and right-shift the smaller significand by the exponent difference), add/subtract (effective operation from the XOR of the signs), normalization (leading-zero count and exponent adjust), rounding (guard and sticky bits per the selected mode, with re-normalization if rounding carries out), and packing with exception flag generation.

Special cases bypass the datapath through a dedicated handler and a final result mux: NaNs propagate quieted with their payload preserved (SoftFloat's default behavior), $\infty - \infty$ returns the canonical qNaN with the invalid flag, infinities dominate, and signed zeros follow the IEEE rules. Overflow saturates to infinity or maximum finite depending on the rounding direction; underflow is flagged only for inexact subnormal results.



Implementation details

Summary

Metric	Value
LUTs	1,141
— normal mode	943
— arith mode	198
Max LUT depth	48.4
Fmax (Slow 1200 mV 85 °C)	30.11 MHz

- ~1,141 LUTs, mostly normal mode (83%). Only 198 cells map to the carry chain (arith mode). The design is dominated by soft combinational logic — the alignment/normalization barrel shifters, the leading-zero counter, and the rounding/mux fabric — rather than adders, which is expected for FP where the mantissa add/subtract is a small fraction of the total.
- Deep critical path (Max LUT depth 48.40). This is the deepest single-cycle block in the project so far, and it directly explains the low Fmax. A binary16 add serializes several variable shifts back-to-back — exponent-difference align → mantissa add/sub → leading-zero normalize shift → round → re-normalize — with no pipeline registers, so the logic levels stack up.
- Fmax 30.11 MHz is the price of the fully combinational, single-cycle design. The obvious lever is pipelining: splitting the datapath into stages (e.g. align / add / normalize+round) would roughly halve the depth and multiply Fmax, at the cost of latency and control — the same trade-off seen in the multiplier and divider analyses.

Timing breakdown

Critical path

All 100 worst-slack paths are at the max depth (38 integer levels / 48.40 weighted), all terminating at *result_q[10]*, launched from 4 different operand bits.

From Node	To Node	Occurrences (of 100)	Logic levels	Total cells
srcb_q[13]	result_q[10]	41	38	95
srcb_q[11]	result_q[10]	24	38	95
srca_q[11]	result_q[10]	19	38	95
srca_q[12]	result_q[10]	16	38	95

These paths share the identical datapath described below.

Logic description	Node	Count
Classify + effective exponent	Equal5/eb_eff/e_small	3–4
Exponent difference	Add0	1
Align — right shift	ShiftRight0	4
Mantissa add/sub	Add1	11
Leading-zero count	f_clz42	6
Shift-amount / exponent math	Add4–Add7	9
Normalize — left shift	ShiftLeft1	3
Exponent-adjust wide add	Add8	31
Rounding decision	round_up	6
Round select mux	Mux1	1
Round increment	Add9	12
Final pack add	Add10	3
Overflow / range compare	LessThan8	2
Output pack	result	2
Total		~95

- The paths collapse to a single end-to-end datapath (classify → align → mantissa add → CLZ → normalize → exponent adjust → round → pack → *result_q[10]*)
- The "different" paths are just the same chain fed from *srca/srcb* bits 11–13.
- *result_q[10]* is the tail because bit 10 is the mantissa/exponent boundary the rounding carry and final mux resolve last.
- Carry chains dominate the depth. *Add8* (31 cells) plus *Add9* (12) account for ~45% of all cells on the path.

Other timing parameters

- **Latency:** 1 cycle (single cycle)
- **Throughput:** 1 operation per cycle
- **Slack:** Significantly negative. Frequency was set to 1 GHz for analysis purposes and getting the maximum possible frequency, which was 30.11 MHz.

Conclusion

The design is functionally bit-exact and area-reasonable, but its single-cycle nature makes it timing-limited, and it is a candidate for width-trimming plus pipelining — the same area/latency/frequency trade-off for pipelined implementations.

The design can be improved to be faster by implementing several classic FP-adder techniques trade area for shorter delay by overlapping stages that are currently serial, for instance: trimming the exponent/shift-amount widths, using a compound-adder for rounding, using a Leading-Zero Anticipator (LZA) to predict the leading-zero count directly from the operands, in parallel with the adder, among other changes.